

Unit 20: Concavity and Curve Sketching — Full Solutions

Warm-up

1. $f(x) = x^3 - 3x^2 + 2$

$$f'(x) = 3x^2 - 6x. \quad f''(x) = 6x - 6. \quad \text{Set } f'' = 0 \Rightarrow x = 1.$$

Test $x < 1$ (0): $f'' = -6 < 0 \rightarrow$ concave down. Test $x > 1$ (2): $f'' = 6 > 0 \rightarrow$ concave up.

Concave down on $(-\infty, 1)$, concave up on $(1, \infty)$. $f(1) = 1 - 3 + 2 = 0$. Inflection point: $(1, 0)$.

2. $f(x) = x^4 - 6x^2$

$$f'(x) = 4x^3 - 12x. \quad f''(x) = 12x^2 - 12 = 12(x-1)(x+1). \quad \text{Critical (for concavity): } x = \pm 1.$$

Test $x < -1$ (-2): using factors: $(-3)(-1) = 3 > 0 \rightarrow$ concave up. Test $-1 < x < 1$ (0): $(-1)(1) = -1 < 0 \rightarrow$ concave down. Test $x > 1$ (2): $(1)(3) = 3 > 0 \rightarrow$ concave up.

Concave up on $(-\infty, -1)$, concave down on $(-1, 1)$, concave up on $(1, \infty)$. $f(-1) = 1 - 6 = -5$, $f(1) = 1 - 6 = -5$. Inflection points: $(-1, -5)$ and $(1, -5)$.

3. $f(x) = x^3 - 6x^2 + 9x$

$$f'(x) = 3x^2 - 12x + 9. \quad f''(x) = 6x - 12. \quad \text{Set } = 0 \Rightarrow x = 2.$$

Test $x < 2$: $f'' < 0 \rightarrow$ concave down. Test $x > 2$: $f'' > 0 \rightarrow$ concave up.

Concave down on $(-\infty, 2)$, concave up on $(2, \infty)$. $f(2) = 8 - 24 + 18 = 2$. Inflection point: $(2, 2)$.

4. $f(x) = -x^3 + 3x$

$$f'(x) = -3x^2 + 3. \quad f''(x) = -6x. \quad \text{Set } = 0 \Rightarrow x = 0.$$

Test $x < 0$: $f'' > 0 \rightarrow$ concave up. Test $x > 0$: $f'' < 0 \rightarrow$ concave down.

Concave up on $(-\infty, 0)$, concave down on $(0, \infty)$. $f(0) = 0$. Inflection point: $(0, 0)$.

5. $f(x) = x^4 - 4x^3 + 1$

$$f'(x) = 4x^3 - 12x^2. \quad f''(x) = 12x^2 - 24x = 12x(x-2). \quad \text{Critical: } x = 0, 2.$$

Test $x < 0$ (-1): $12(-1)(-3) = 36 > 0 \rightarrow$ concave up. Test $0 < x < 2$ (1): $12(1)(-1) = -12 < 0 \rightarrow$ concave down. Test $x > 2$ (3): $12(3)(1) = 36 > 0 \rightarrow$ concave up.

Concave up on $(-\infty, 0)$, concave down on $(0, 2)$, concave up on $(2, \infty)$. $f(0) = 1$, $f(2) = 16 - 32 + 1 = -15$. Inflection points: $(0, 1)$ and $(2, -15)$.

6. $f(x) = x^3 - 3x + 1$

$$f'(x) = 3x^2 - 3. \quad f''(x) = 6x. \quad \text{Set } = 0 \Rightarrow x = 0.$$

Test $x < 0$: $f'' < 0 \rightarrow$ concave down. Test $x > 0$: $f'' > 0 \rightarrow$ concave up.

Concave down on $(-\infty, 0)$, concave up on $(0, \infty)$. $f(0) = 1$. Inflection point: $(0, 1)$.

Standard

7. $f(x) = x^3 - 6x^2 + 9x + 1$

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. Critical points: $x = 1, 3$. $f''(x) = 6x - 12$.

$f''(1) = 6 - 12 = -6 < 0 \rightarrow$ local max at $x = 1$, $f(1) = 1 - 6 + 9 + 1 = 5$.

$f''(3) = 18 - 12 = 6 > 0 \rightarrow$ local min at $x = 3$, $f(3) = 27 - 54 + 27 + 1 = 1$.

8. $f(x) = x^4 - 4x^3 + 2$

$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$. Critical points: $x = 0$ (double), $x = 3$. $f''(x) = 12x^2 - 24x = 12x(x-2)$.

$f''(0) = 0 \rightarrow$ inconclusive, must use the First Derivative Test. $f'(x) = 4x^2(x-3)$: test $x < 0$ (-1): $4(-1)(-4) = -16 < 0$; test $0 < x < 3$ (1): $4(1)(-2) = -8 < 0$. Same sign both sides $\rightarrow x = 0$ is neither a local max nor min, $f(0) = 2$.

$f''(3) = 12(3)(1) = 36 > 0 \rightarrow$ local min at $x = 3$, $f(3) = 81 - 108 + 2 = -25$.

9. $f(x) = 2x^3 - 3x^2 - 12x + 5$

$f'(x) = 6x^2 - 6x - 12$. $f''(x) = 12x - 6$. Set $= 0 \Rightarrow x = \frac{1}{2}$.

Test $x < \frac{1}{2}$: $f'' < 0 \rightarrow$ concave down. Test $x > \frac{1}{2}$: $f'' > 0 \rightarrow$ concave up.

Concave down on $(-\infty, \frac{1}{2})$, concave up on $(\frac{1}{2}, \infty)$. $f(\frac{1}{2}) = 2(\frac{1}{8}) - 3(\frac{1}{4}) - 12(\frac{1}{2}) + 5 = 0.25 - 0.75 - 6 + 5 = -1.5$. Inflection point: $(\frac{1}{2}, -1.5)$.

10. $f(x) = x^4 - 2x^2 + 3$

$f'(x) = 4x^3 - 4x$. $f''(x) = 12x^2 - 4 = 4(3x^2 - 1)$. Set $= 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{\sqrt{3}}{3}$.

Test $x < -\frac{\sqrt{3}}{3}$ (-1): $4(3-1) = 8 > 0 \rightarrow$ concave up. Test $-\frac{\sqrt{3}}{3} < x < \frac{\sqrt{3}}{3}$ (0): $4(0-1) = -4 < 0 \rightarrow$ concave down. Test $x > \frac{\sqrt{3}}{3}$ (1): $4(3-1) = 8 > 0 \rightarrow$ concave up.

Concave up on $(-\infty, -\frac{\sqrt{3}}{3})$, concave down on $(-\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3})$, concave up on $(\frac{\sqrt{3}}{3}, \infty)$.

$f(\frac{\sqrt{3}}{3}) = (\frac{1}{3})^2 - 2(\frac{1}{3}) + 3 = \frac{1}{9} - \frac{6}{9} + \frac{27}{9} = \frac{22}{9}$. By symmetry (even function), the same value occurs at $x = -\frac{\sqrt{3}}{3}$.

Inflection points: $\left(\pm \frac{\sqrt{3}}{3}, \frac{22}{9}\right)$.

11. $f(x) = x^3 - 3x^2 - 9x + 1$

$f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1)$. Critical points: $x = 3, -1$. $f''(x) = 6x - 6$.

$f''(-1) = -6 - 6 = -12 < 0 \rightarrow$ local max at $x = -1$, $f(-1) = -1 - 3 + 9 + 1 = 6$.

$f''(3) = 18 - 6 = 12 > 0 \rightarrow$ local min at $x = 3$, $f(3) = 27 - 27 - 27 + 1 = -26$.

12. $f(x) = 3x^5 - 5x^3$

$f'(x) = 15x^4 - 15x^2$. $f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$. Set $= 0 \Rightarrow x = 0$ or $x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{\sqrt{2}}{2}$.

Test $x < -\frac{\sqrt{2}}{2}$ (-1): $30(-1)(2-1) = -30 < 0 \rightarrow$ concave down.

Test $-\frac{\sqrt{2}}{2} < x < 0$ (-0.5): $30(-0.5)(0.5-1) = 7.5 > 0 \rightarrow$ concave up.

Test $0 < x < \frac{\sqrt{2}}{2}$ (0.5): $30(0.5)(0.5-1) = -7.5 < 0 \rightarrow$ concave down.

Test $x > \frac{\sqrt{2}}{2}$ (1): $30(1)(2-1) = 30 > 0 \rightarrow$ concave up.

Concave down on $(-\infty, -\frac{\sqrt{2}}{2})$, concave up on $(-\frac{\sqrt{2}}{2}, 0)$, concave down on $(0, \frac{\sqrt{2}}{2})$, concave up on $(\frac{\sqrt{2}}{2}, \infty)$.

$f(0) = 0$. $f(\frac{\sqrt{2}}{2}) = 3(\frac{\sqrt{2}}{8}) - 5(\frac{\sqrt{2}}{4}) = \frac{3\sqrt{2}}{8} - \frac{10\sqrt{2}}{8} = -\frac{7\sqrt{2}}{8}$. Since f is an odd function, $f(-\frac{\sqrt{2}}{2}) = \frac{7\sqrt{2}}{8}$.

Inflection points: $(-\frac{\sqrt{2}}{2}, \frac{7\sqrt{2}}{8})$, $(0, 0)$, and $(\frac{\sqrt{2}}{2}, -\frac{7\sqrt{2}}{8})$.

Challenge

13. $f(x) = -x^3 + 6x^2 - 9x + 3$

Domain: all real numbers. Asymptotes: none (a polynomial never has any).

$f'(x) = -3x^2 + 12x - 9 = -3(x-1)(x-3)$. Critical points: $x = 1, 3$.

Test $x < 1$ (0): $-3(-1)(-3) = -9 < 0 \rightarrow$ decreasing. Test $1 < x < 3$ (2): $-3(1)(-1) = 3 > 0 \rightarrow$ increasing. Test $x > 3$ (4): $-3(3)(1) = -9 < 0 \rightarrow$ decreasing.

Decreasing on $(-\infty, 1)$, increasing on $(1, 3)$, decreasing on $(3, \infty)$.

$f''(x) = -6x + 12$. $f''(1) = 6 > 0 \rightarrow$ local min at $x = 1$, $f(1) = -1 + 6 - 9 + 3 = -1$. $f''(3) = -6 < 0 \rightarrow$ local max at $x = 3$, $f(3) = -27 + 54 - 27 + 3 = 3$.

Set $f'' = 0 \Rightarrow x = 2$. Test $x < 2$: $f'' > 0 \rightarrow$ concave up. Test $x > 2$: $f'' < 0 \rightarrow$ concave down. $f(2) = -8 + 24 - 18 + 3 = 1$.

Concave up on $(-\infty, 2)$, concave down on $(2, \infty)$. Inflection point: $(2, 1)$.

No absolute extrema — as $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$ (the leading $-x^3$ term dominates in both directions).

Sketch summary: the curve comes down from $+\infty$ on the far left, decreases to a local minimum at $(1, -1)$, rises to a local maximum at $(3, 3)$, then decreases forever toward $-\infty$. It's concave up throughout the first half (up to $x = 2$) and concave down after — the inflection point $(2, 1)$ sits right between the local min and local max.

14. $f(x) = \frac{(x+1)^2}{1+x^2}$

Domain: all real numbers (denominator $1+x^2$ is never 0).

Asymptotes: numerator and denominator both degree 2 with leading coefficient 1, so horizontal asymptote $y = 1$. No vertical asymptotes.

Using the quotient rule (with $N = (x+1)^2$, $N' = 2(x+1)$; $D = 1+x^2$, $D' = 2x$), and simplifying (factoring $2(x+1)$ out of the numerator):

$$f'(x) = \frac{2(x+1)(1+x^2) - (x+1)^2(2x)}{(1+x^2)^2} = \frac{2(x+1)(1-x)}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2}$$

Critical points: numerator zero at $x = \pm 1$; denominator never zero.

Test $x < -1$ (−2): $2(1-4) = -6 < 0 \rightarrow$ decreasing. Test $-1 < x < 1$ (0): $2(1) = 2 > 0 \rightarrow$ increasing. Test $x > 1$ (2): $2(1-4) = -6 < 0 \rightarrow$ decreasing.

Decreasing on $(-\infty, -1)$, increasing on $(-1, 1)$, decreasing on $(1, \infty)$.

$$f(-1) = \frac{0}{2} = 0 \rightarrow \text{local min. } f(1) = \frac{4}{2} = 2 \rightarrow \text{local max.}$$

For concavity, differentiate $f'(x) = \frac{2(1-x^2)}{(1+x^2)^2}$ again (quotient rule, then simplify — the algebra is involved but the common factor $4x(1+x^2)$ cancels a layer):

$$f''(x) = \frac{-4x(3-x^2)}{(1+x^2)^3} = \frac{4x(x^2-3)}{(1+x^2)^3}$$

Critical points for concavity: $x = 0$ or $x^2 = 3 \Rightarrow x = \pm\sqrt{3}$ (denominator always positive).

Test $x < -\sqrt{3}$ (−2): $4(-2)(4-3) = -8 < 0 \rightarrow$ concave down. Test $-\sqrt{3} < x < 0$ (−1): $4(-1)(1-3) = 8 > 0 \rightarrow$ concave up. Test $0 < x < \sqrt{3}$ (1): $4(1)(1-3) = -8 < 0 \rightarrow$ concave down. Test $x > \sqrt{3}$ (2): $4(2)(4-3) = 8 > 0 \rightarrow$ concave up.

Concave down on $(-\infty, -\sqrt{3})$, concave up on $(-\sqrt{3}, 0)$, concave down on $(0, \sqrt{3})$, concave up on $(\sqrt{3}, \infty)$.

$$f(0) = 1. \quad f(\sqrt{3}) = \frac{(\sqrt{3}+1)^2}{4} = \frac{4+2\sqrt{3}}{4} = 1 + \frac{\sqrt{3}}{2}. \quad f(-\sqrt{3}) = \frac{(1-\sqrt{3})^2}{4} = \frac{4-2\sqrt{3}}{4} = 1 - \frac{\sqrt{3}}{2}.$$

Inflection points: $\left(-\sqrt{3}, 1 - \frac{\sqrt{3}}{2}\right)$, $(0, 1)$, and $\left(\sqrt{3}, 1 + \frac{\sqrt{3}}{2}\right)$.

Absolute extrema: since $f(x) = \frac{(x+1)^2}{1+x^2} \geq 0$ always (a square divided by a positive number), and $f(-1) = 0$ is the smallest possible value, the local min at $x = -1$ ($f = 0$) is also the absolute minimum. Since $f \rightarrow 1$ at $\pm\infty$ and the only other critical value is $f(1) = 2 > 1$, the local max at $x = 1$ ($f = 2$) is also the absolute maximum.

$$15. \quad f(x) = x^4 - 4x^3 + 4x^2 = x^2(x-2)^2$$

Domain: all real numbers. Asymptotes: none.

$$f'(x) = 4x^3 - 12x^2 + 8x = 4x(x-1)(x-2). \quad \text{Critical points: } x = 0, 1, 2.$$

$$f''(x) = 12x^2 - 24x + 8 = 4(3x^2 - 6x + 2).$$

$$f''(0) = 8 > 0 \rightarrow \text{local min at } x = 0, \quad f(0) = 0.$$

$$f''(1) = 4(3-6+2) = -4 < 0 \rightarrow \text{local max at } x = 1, \quad f(1) = 1-4+4 = 1.$$

$$f''(2) = 4(12-12+2) = 8 > 0 \rightarrow \text{local min at } x = 2, \quad f(2) = 16-32+16 = 0.$$

Increasing on $(0, 1) \cup (2, \infty)$, decreasing on $(-\infty, 0) \cup (1, 2)$ (consistent with the sign-change pattern of $4x(x-1)(x-2)$).

$$\text{Set } f'' = 0: 3x^2 - 6x + 2 = 0 \Rightarrow x = \frac{6 \pm \sqrt{36-24}}{6} = \frac{6 \pm 2\sqrt{3}}{6} = 1 \pm \frac{\sqrt{3}}{3}.$$

Testing around these values (using $f''(0) = 8 > 0$, $f''(1) = -4 < 0$, $f''(2) = 8 > 0$ as reference points, since $0 < 1 - \frac{\sqrt{3}}{3} < 1 < 1 + \frac{\sqrt{3}}{3} < 2$):

Concave up on $(-\infty, 1 - \frac{\sqrt{3}}{3})$, concave down on $(1 - \frac{\sqrt{3}}{3}, 1 + \frac{\sqrt{3}}{3})$, concave up on $(1 + \frac{\sqrt{3}}{3}, \infty)$.

By the symmetry $f(x) = ((x-1)^2 - 1)^2$ around $x = 1$, both inflection points share the same height: $f(1 \pm \frac{\sqrt{3}}{3}) = (\frac{1}{3} - 1)^2 = (-\frac{2}{3})^2 = \frac{4}{9}$.

Inflection points: $(1 - \frac{\sqrt{3}}{3}, \frac{4}{9})$ and $(1 + \frac{\sqrt{3}}{3}, \frac{4}{9})$.

16. $f(x) = (x-2)^3$

Domain: all real numbers. Asymptotes: none.

$f'(x) = 3(x-2)^2$. Critical point: $x = 2$ (the only place $f' = 0$; note $f'(x) \geq 0$ everywhere, since it's a square times a positive constant).

Since $f'(x) \geq 0$ on both sides of $x = 2$ (never negative), the sign of f' does not change — $x = 2$ is neither a local max nor a local min. f is increasing everywhere except momentarily flat at $x = 2$.

$$f''(x) = 6(x-2). \text{ Set } = 0 \Rightarrow x = 2.$$

Test $x < 2$: $f'' < 0 \rightarrow$ concave down. Test $x > 2$: $f'' > 0 \rightarrow$ concave up.

Concave down on $(-\infty, 2)$, concave up on $(2, \infty)$. $f(2) = 0$. Inflection point: $(2, 0)$.

This is a nice example of a point that is simultaneously a critical point (flat tangent), an inflection point (concavity changes), and yet not a local extremum at all — a horizontal “saddle” point where the graph pauses momentarily but keeps climbing in the same direction.

17. $f(x) = \frac{1}{x^2 + 1}$

Domain: all real numbers (denominator never 0). Asymptotes: horizontal asymptote $y = 0$ (top degree < bottom degree); no vertical asymptotes.

$$f'(x) = -\frac{2x}{(x^2 + 1)^2}. \text{ Critical point: } x = 0 \text{ (numerator zero; denominator never zero).}$$

Test $x < 0$: $f' > 0 \rightarrow$ increasing. Test $x > 0$: $f' < 0 \rightarrow$ decreasing.

Increasing on $(-\infty, 0)$, decreasing on $(0, \infty)$. Local max at $x = 0$, $f(0) = 1$.

$$f''(x) = \frac{6x^2 - 2}{(x^2 + 1)^3}$$

$$\text{Set numerator} = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{\sqrt{3}}{3}.$$

Test $x < -\frac{\sqrt{3}}{3}$ (-1): $6 - 2 = 4 > 0 \rightarrow$ concave up. Test $-\frac{\sqrt{3}}{3} < x < \frac{\sqrt{3}}{3}$ (0): $-2 < 0 \rightarrow$ concave down.
 Test $x > \frac{\sqrt{3}}{3}$ (1): $4 > 0 \rightarrow$ concave up.

Concave up on $(-\infty, -\frac{\sqrt{3}}{3})$, concave down on $(-\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3})$, concave up on $(\frac{\sqrt{3}}{3}, \infty)$.

$f(\frac{\sqrt{3}}{3}) = \frac{1}{\frac{1}{3} + 1} = \frac{3}{4}$. By symmetry, the same value occurs at $x = -\frac{\sqrt{3}}{3}$.

Inflection points: $(\pm \frac{\sqrt{3}}{3}, \frac{3}{4})$.

Absolute max: 1 at $x = 0$ (since $f(x) \leq 1$ always). No absolute minimum — $f(x)$ approaches 0 as $x \rightarrow \pm\infty$ but never actually reaches it.

Applied

18. $C(x) = x^3 - 9x^2 + 30x + 50$

$C'(x) = 3x^2 - 18x + 30$. $C''(x) = 6x - 18$. Set $= 0 \Rightarrow x = 3$.

Test $x < 3$: $C'' < 0 \rightarrow$ concave down. Test $x > 3$: $C'' > 0 \rightarrow$ concave up.

$C(3) = 27 - 81 + 90 + 50 = 86$.

Inflection point: $(3, 86)$. This is the production level where the cost curve switches from concave down (marginal cost still rising, but at a slowing rate — a period of relative efficiency) to concave up (marginal cost now rising at an accelerating rate — diseconomies of scale taking hold).

19. $N(t) = -t^3 + 15t^2 + 100$

$N'(t) = -3t^2 + 30t$. $N''(t) = -6t + 30$. Set $= 0 \Rightarrow t = 5$.

Test $t < 5$: $N'' > 0 \rightarrow$ concave up. Test $t > 5$: $N'' < 0 \rightarrow$ concave down.

$N(5) = -125 + 375 + 100 = 350$.

Inflection point: $(5, 350)$. This marks the moment of fastest growth (the steepest point on the curve). Before $t = 5$, the user count is growing at an accelerating rate; after $t = 5$, it's still growing, but the rate of growth itself starts to slow down — the classic shape of an S-curve's steepest crossing point.

20. $s(t) = t^3 - 6t^2 + 9t$

$s'(t) = 3t^2 - 12t + 9$. $s''(t) = 6t - 12$. Set $= 0 \Rightarrow t = 2$.

Test $t < 2$: $s'' < 0 \rightarrow$ concave down. Test $t > 2$: $s'' > 0 \rightarrow$ concave up.

$s(2) = 8 - 24 + 18 = 2$.

Inflection point: $(2, 2)$. Since $a(t) = s''(t)$ is the object's acceleration, this inflection point is exactly the moment the acceleration changes sign — from negative (before $t = 2$) to positive (after $t = 2$). This connects directly to the motion analysis from Unit 11: it's the pivot moment in the object's "story," where the pattern of speeding up and slowing down flips over.

21. $P(x) = -x^3 + 12x^2 - 20x$

$$P'(x) = -3x^2 + 24x - 20. \quad P''(x) = -6x + 24. \quad \text{Set } = 0 \Rightarrow x = 4.$$

Test $x < 4$: $P'' > 0 \rightarrow$ concave up. Test $x > 4$: $P'' < 0 \rightarrow$ concave down.

$$P(4) = -64 + 192 - 80 = 48.$$

Inflection point: $(4, 48)$. This marks the production level where returns begin to diminish: before $x = 4$, marginal profit (the slope of the profit curve) is increasing — each additional unit is adding more profit than the one before. After $x = 4$, marginal profit starts decreasing — profit is still growing (up to a point), but each additional unit contributes less than the last.